

Project 9:- Blockchain-Based Academic Certificate Verification

Verification Results Report

Dataset Overview

- Certificates tested: 5
- Universities: SNDTW University, Pune University, Mumbai University
- Fields used for hashing: Student ID, Certificate ID, Award Date

Verification Method

Each certificate's Student ID, Certificate ID, and Award Date were concatenated and hashed using SHA-256 to produce a unique digital fingerprint. These hashes were stored immutably on the Ethereum test network via a Solidity smart contract.

During verification, the uploaded certificate was re-hashed and compared against the blockchain entry. A matching hash confirmed authenticity, while a mismatch indicated tampering.

Results

Student Name	Certificate ID	Hash Match	Status
Bhargavi Kambli	CERTO08	✓	Authentic
Vaishnavi Mehatri	CERTO18	✓	Authentic
Neha Pawar	CERTO04	✗	Tampered
Vanshika Foshi	CERT001	✓	Authentic
Riya Patel	CERTO69	✓	Authentic

Summary & Insights

- Total certificates tested: 5
- Authentic: 4
- Tampered/Invalid: 1
- Observation: Blockchain verification successfully detected inconsistencies in certificate CERT004.

Conclusion

Blockchain hashing ensures tamper-proof verification of academic certificates. The system reliably distinguishes authentic records from altered ones. Future work may include integration with university portals and scaling to larger datasets.